

KYC Verification Waterfall v2 - ARD

Proving the waterfall works. Twenty metrics in four tiers, each with a numerator, a denominator, a baseline and an alert threshold - three of them invariants, not KPIs, plus the guardrails that prove we did not buy the rate by weakening a control.

This document is self-contained. §1 gives the system context an analyst needs; §2 specifies the instrumentation everything else reads from; §3 defines every metric with an explicit numerator, denominator, baseline and threshold. Baselines are measured over 2,515,155 applications enrolled Jan 2023 – May 2026, with 107 internal test accounts excluded.

1. What we are measuring, and why

The system. Every applicant for our US consumer product must clear identity verification (KYC), which runs as a waterfall: a primary provider (**Idology**), then secondary providers (**LexisNexis** and **Persona-SSN** on the SSN path; **ACRO** and **Persona IDV** on the non-SSN path), then **manual review** by a human analyst. The field recording the final outcome is `overall_kyc_status`.

The problem v2 fixes. 7.93% of applicants end up not verified against a 5% bar. The dominant cause is not that people fail checks — it is that the waterfall stops running:

Defect	Volume	Verified
Stranded — primary check failed, nothing else ever ran	101,162 (4.02% of book, 50.7% of all non-verifications)	0.003%
Primary bypassed — Idology never ran	94,183 (3.74%)	99.08%
PASS did not exit — clean PASS, further checks ran anyway	131,538 (5.23%)	99.81% (247 lost)
Safety net missed — automation exhausted, no human review	48,598 non-verified	0%

Comparable applicants who *were* routed onward after a primary failure recovered at **48.09%**. 75.6% of all non-verification is process failure rather than a correct decision.

What v2 changes. Every primary-check failure is classified into one reason (`SANCTIONS_HIT` / `SSN_MISMATCH` / `IDENTITY_ATTR_MISMATCH` / `NO_DATA`), every reason has a defined path, a clean PASS terminates the flow unconditionally, and every application ends in exactly one of three terminal states: `VERIFIED` , `HARD_REJECT` , `IN_MANUAL_REVIEW` . No verification standard or threshold changes. Full design in the companion **PRD**.

What this plan must prove. Three things, in order: (1) the waterfall's structural promises hold — these are invariants, not KPIs; (2) the recovery is real and the routing is sane; (3) **we did not buy the rate by weakening a control**.

2. Instrumentation

2.1 The event log

Every metric below reads from one new **append-only** table, `kyc_decision_event` — one row per check attempt and per routing decision:

Column	Type	Notes
event_id	uuid	Primary key
application_id	uuid	Grain for all rate metrics
attempt_seq	int	Monotonic within an application
tier	enum	L1 L1.5 L2a L2b L2 L3 L4
provider	enum	IDOLGY LEXISNEXIS PERSONA_SSN ACRO PERSONA_IDV MANUAL NONE
reason_class	enum	SANCTIONS_HIT SSN_MISMATCH IDENTITY_ATTR_MISMATCH NO_DATA NULL
decision	enum	PASS FAIL ERROR TIMEOUT SKIPPED
routed_to	enum	Next tier, or TERMINAL
terminal_state	enum	VERIFIED HARD_REJECT IN_MANUAL_REVIEW NULL
latency_ms	int	Provider round-trip
cost_cents	int	Populated from the contracted rate card
rollout_cohort	enum	TREATMENT CONTROL SHADOW
flag_state	jsonb	Which feature flags were on for this application
event_ts	timestampz	

Plus a satellite table `kyc_idv_funnel` — one row per IDV `invitation`, `start`, `completion`, `abandonment` — because an abandoned document-and-selfie flow currently reads as "not verified" and is indistinguishable from a decline.

This is not optional plumbing. The baseline analysis could only reconstruct paths by *inferring* them from per-provider result columns, and the `kyc_source` field turned out not to be a substitute — it names one credited clearing provider per applicant, not every check that ran. v2 must not be auditable only by inference.

2.2 Rules that apply to every metric

1. **Grain is the application, not the check.** An applicant is counted once per application.
2. **Exclude internal test accounts** (`is_test_user`) from every numerator and denominator, always.
3. **Cohort every rate.** Treatment vs. control, from the same time window, using the sticky application-id hash. Absolute rates without a cohort split are not decision-grade during rollout.
4. **Completed applications only** for outcome metrics — an application still inside its retry window is neither verified nor rejected. Maturity window: 72h for automated paths, 14d where L4 is involved.
5. **State the denominator in the metric name.** Every definition below names both sides of the ratio.

3. Metrics

Tier 1 — do the design's structural promises hold?

#	Metric	Definition (numerator ÷ denominator)	Baseline	Target	Alert
M1	Non-verification rate	applications not <code>VERIFIED</code> ÷ all completed applications	7.93%	≤6.3%; 5.0% stretch	P2: >6.5% sustained 7d
M2	Stranded rate	applications with ≥1 <code>FAIL</code> , no pending check and no terminal state ÷ all applications	4.02% (101,162)	0	P1: >0.1% over 1h
M3	L1 coverage	applications with an <code>L1</code> event ÷ all applications	96.26%	100%	P1: <99.9% over 1h
M3b	Ex-bypass cohort verification	<code>VERIFIED</code> ÷ applications that would have skipped L1 under the old rules	99.08%	≥97%	P2: <95%
M4	PASS-exit integrity	clean-PASS applications with ≥1 downstream event ÷ clean-PASS applications	5.23% (131,538)	0	P1: >0.05% over 1h
M5	Manual-review coverage	applications with an <code>L4</code> event ÷ applications that exhausted their automated path	30.2% (64,414 of 213,048)	100%	P1: <99% over 24h
M6	Terminal-state completeness	applications in exactly one terminal state at 72h ÷ all applications	not measurable pre-v2	100%	P1: <99.95%

M2, M4 and M6 are invariant checks, not KPIs. A non-zero M2 or M4, or an M6 below 100%, is a **bug**, not a bad week. They alert on any breach, not on a trend, and they map directly to the PRD's FR-1 / FR-2 / FR-3 assertions.

Tier 2 — is the routing sane?

#	Metric	Definition	Baseline	Target	Alert
M7	Sanctions routing SLA	<code>SANCTIONS_HIT</code> applications reaching the compliance queue ÷ all such applications; and time-to-queue	no distinct route today	100%, p95 ≤4h	P1 on any miss
M8	<code>NO_DATA</code> share	L1 <code>FAILs</code> classed <code>NO_DATA</code> ÷ all L1 <code>FAILs</code>	n/a	≤10%	P2: >20% for 24h
M8b	Classifier agreement	classifier class matching the reviewer-assigned class ÷ L4 cases with a reviewer class	n/a	≥90%	P2: <85% weekly
M9	Fallback recovery rate	<code>VERIFIED</code> ÷ applications entering any L2/L3 tier	48.09% observed	≥45%	P2: <35% for 7d
M9b	Crossover recovery rate	<code>VERIFIED</code> ÷ applications taking the SSN→non-SSN crossover	55.0% (8,605 cases)	≥50%	P3
M9c	IDV effectiveness	<code>VERIFIED</code> ÷ applications reaching L3	86.3% (on 2,507 cases, 0.87% of failures)	≥70% at volume	P2: <60%
M10	Per-tier pass rate	PASS ÷ attempts, by provider, daily	per-provider trailing	within ±5pp of 28d trailing	P2 on drift

M9c deserves scepticism, not celebration. 86.3% is measured on 0.87% of eligible volume — an unrepresentative, probably self-selected sample. Expect regression at 10–20× volume. Its target is set well below its baseline for exactly this reason.

Tier 3 — guardrails: did we weaken a control?

#	Metric	Definition	Baseline	Target	Alert
G-a	Manual-review decline rate	<code>HARD_REJECT</code> ÷ L4 cases decided	pre-rollout trailing	within ±5pp	P2 on breach
G-b	Sanctions true-hit rate	confirmed hits ÷ <code>SANCTIONS_HIT</code> cases	≈21% (79.0% of 9,994 cleared as false positives)	±5pp	P1 on a sharp drop — implies over-firing or mis-routing
G-c	Post-verification fraud / chargeback rate	fraud-flagged ÷ verified, 90-day cohort	current trailing	no increase vs. control	P1 on any increase

G-c is the metric that decides whether v2 was actually a good idea. Every other metric can read green while we verify people we should not have. It lags 90 days — which is precisely why the control cohort must survive to P4. Cut over to 100% early and the comparison is gone permanently.

Tier 4 — cost, latency, friction

#	Metric	Definition	Baseline	Target	Alert
C1	Provider spend per verified applicant	$\Sigma \text{cost_cents} \div \text{VERIFIED}$	illustrative	≤ +20% vs. baseline	P2 on breach
C2	L4 case volume	L4 cases per week	~21,900/yr (74,400 cases over the window)	≤ modelled 2.4× (~51,600/yr)	P1: >3.0× — capacity breach
C3	L4 turnaround	queue-entry → decision	current trailing	p50 <24h, p90 <72h	P1: p90 >120h
C4	IDV completion	completions ÷ invitations (<code>kyc_idv_funnel</code>)	unmeasurable today	≥70%	P2: <60%
C5	End-to-end decision latency	submit → terminal state	p50 <2s	p50 unchanged; p90 <5s excl. L3/L4	P2 on p50 regression

C4 is the friction metric we currently cannot see at all. A 10–20× increase in IDV volume into a 60–120 second document-and-selfie flow is a real abandonment risk, and today an abandoned IDV is recorded identically to a decline. Without the separate invitation/start/completion instrumentation, C4 is unmeasurable and we will misread drop-off as rejection.

4. Measurement design

Assignment. Sticky hash of `application_id`, fixed at first submission. Applications never change cohort mid-flight, so no applicant experiences two routing regimes.

Primary readout. Difference in M1 between treatment and control, on applications *submitted* within the phase window and matured to the 14-day boundary. Report the absolute difference in percentage points with a 95% interval, never a relative "x% better".

Sample adequacy. At 5% traffic (P1) the cohort is too small to resolve a ~1.9pp move in M1 within two weeks — P1 is a **correctness** phase, and its exit gates are invariants (M2, M6) and queue health (C3), not conversion. M1 becomes readable at P2 (25%) and decision-grade at P3 (50%).

Three effects must be reported separately, never netted:

Effect	Why it must be isolated
Routing fix (the stranded population)	The genuine win. This is what "did v2 work?" means.
Bypass retirement (M3b)	Expected to <i>reduce</i> verification rate — it forces 94,183 applicants/window who currently verify at 99.08% through a real primary check. Bundled into M1, it hides a real win under a control improvement.
Stranded backlog migration	A one-off recovery batch, not steady-state behaviour. Its recoveries are excluded from the M1 series, or the running system looks better than it is.

Attribution. With per-change feature flags (PRD FR-11), `flag_state` on every event allows M1 to be decomposed by flag combination. Analysts should decompose before attributing any movement.

5. Dashboards

Dashboard	Audience	Refresh	Contents
Funnel Health	Exec / Product	Daily	M1 against the 5% bar, treatment vs. control; verification by tier; week-over-week trend; the A–D process-failure bucket mix restated on live data
Routing Integrity	Backend Eng	5 min	M2, M3, M4, M5, M6 as invariant panels — green/red, not trend lines ; per-tier volume; error and timeout rates
Ops Capacity	Review Ops	Hourly	C2 queue depth, C3 turnaround by percentile, arrival vs. completion rate, backlog burn-down, analyst utilisation, compliance queue shown separately from identity review
Provider & Cost	Product / Finance / Vendor mgmt	Weekly	C1, per-provider volume · pass rate · latency · spend, C4 IDV funnel, cost per incremental verification
Rollout Scorecard	Everyone, P0–P4	Daily	Every phase-gate metric, treatment vs. control side by side, with the gate threshold drawn on each chart

The Rollout Scorecard exists so each phase-gate decision is made against a **pre-agreed panel** rather than re-argued from scratch. It is retired at P4.

6. Alerts

Sev	Trigger	Route	Response
P1	An invariant is broken or a control is at risk — M2, M3, M4, M5, M6, M7, G-b, G-c, C2, C3	Page on-call engineer; Compliance also paged for M7 / G-b	Assess within 15 min. Any unresolved P1 for >1h auto-halts the ramp — freeze at the current percentage, do not auto-revert.
P2	A KPI or trade-off is off target — M1, M3b, M8, M8b, M9, M9c, M10, C1, C4, G-a	Slack to Product + Eng + Ops	Triage same business day; blocks the next phase gate until explained
P3	Informational drift — M9b, minor provider variance	Weekly digest	Reviewed at the weekly rollout stand-up

Why halt rather than auto-revert: flipping routing mid-application is its own failure mode. Rollback is a human decision, with the legacy path kept warm through P4.

Two alerts that must exist and are easy to forget:

- **Silence alert.** If `kyc_decision_event` volume for any tier drops >50% hour-over-hour, page. A tier that has stopped firing looks identical to a tier with nothing to do — and the stranding defect went unnoticed for three and a half years precisely because *absence generates no signal*.
- **Reconciliation alert.** Daily comparison of v2 terminal states against the legacy result columns (both are written through P4). Any divergence outside the expected treatment effect is an integration bug, not a product result.

7. What we watch in each phase

Phase	The question being answered	Watch	Stop / do not proceed if
P0 Shadow (100% mirrored, 0% acting)	Would v2 have decided differently, and can the classifier be trusted?	M8b agreement on the reviewer-labelled subset; projected L4 volume vs. model; shadow M2 / M4	Agreement <95%, or projected L4 volume >1.2× model
P1 Canary (5%)	Does it work at all, and does the queue hold?	M2 (must be 0), M6, C3, every P1 alert	Any P1 alert; C3 p90 >72h
P2 Ramp (25%)	Is the recovery real, and what does IDV friction actually cost?	M1 in cohort, M9, M9c, C4 , C1	M1 not improving vs. control; C4 <60%
P3 Majority (50%)	Are the guardrails holding, and is the cost defensible?	G-a, G-b, early G-c signal, C1, C2	G-a moves >5pp; G-c shows any fraud increase
P4 Full (100%)	Did we hit the goal, and can we prove it?	M1 full population, re-baselined target, 90-day G-c readout	—

Two things are deliberately not ramped with the main flag and are measured on their own series: bypass retirement (M3b, expected to cost rate) and the stranded-backlog migration (a one-off, excluded from the steady-state M1 series).

8. Honest limits of this plan

- **The 48.09% recovery figure is a comparison-group estimate, not a measured outcome.** Stranded applicants never received a second check, so their true recoverability is unobserved; the comparison group may be an easier population. P1 and P2 exist to measure it. If M9 lands materially below 45%, the $\leq 6.3\%$ target is **wrong and gets re-baselined at P3** — stated now, not discovered late.
- **M9c's 86.3% IDV effectiveness rests on 2,507 cases, 0.87% of eligible volume.** Treat it as a hypothesis to be tested at P2, not an input to a forecast.
- **All C1/C2 cost figures use illustrative unit costs applied to real volumes.** The volumes are load-bearing; the unit costs are not. Contracted rates go in before any threshold is treated as a budget commitment.
- **G-c lags 90 days.** For its first quarter, v2 is judged on process-integrity metrics (Tier 1) plus the best available proxy — manual-review decline-rate stability (G-a). We should say plainly that the outcome question is open until the first fraud cohort matures.
- **M8b needs ground truth that does not exist yet.** Classifier agreement is measured against reviewer-assigned classes, which requires reviewers to record a class on every L4 case from P0 onward. If that field is not populated, M8b is unmeasurable and the P0 exit gate cannot be evaluated.